



WATCHDOG ENERGY MANAGEMENT

Customer User Manual

Version 4.2.0

Watch Every Watt

Electricity and water monitoring, billing and invoice management

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About this manual

This manual explains how customers use Watchdog Energy Management version 4.2.0. It is written for mall managers, billing staff, operators, administrators and viewers. It describes only functions present in this release.

Demonstration data

All screenshots use fictional shops, names, email addresses, meters, readings and invoice values. They do not contain customer credentials or private operational data.

Document conventions

Field or status	Meaning
Bold text	A visible button, menu name, field, status or important result.
Administrator only	The action requires an account assigned the Administrator access level.
Optional feature	The page appears only after an Administrator enables it.
Main Watchdog PC	The Windows computer running the Watchdog server, database and meter polling services.
Browser client	Another computer or mobile device that views the same system over the local network.

System responsibility

Watchdog stores the cumulative values supplied by the physical meter, DDC, PLC, gateway or BACnet device. It calculates consumption as the difference between valid cumulative readings and applies the meter-specific tariff. It does not replace the physical meter's energy or volume calculation.

Important

Verify every register, BACnet point, engineering unit, scale and tariff against the meter or gateway documentation before using readings for billing.

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1. Quick start

Start Watchdog

- 1 On the main Windows PC, double-click the **Watchdog Energy Management** desktop shortcut.
- 2 Wait for the start screen to close and the browser to open.
- 3 If the browser does not open automatically, open **http://localhost:5080** on the main PC.
- 4 Sign in with the username and password supplied by your Administrator.
- 5 Confirm that the Dashboard loads and that meter statuses begin updating.

Recommended daily check

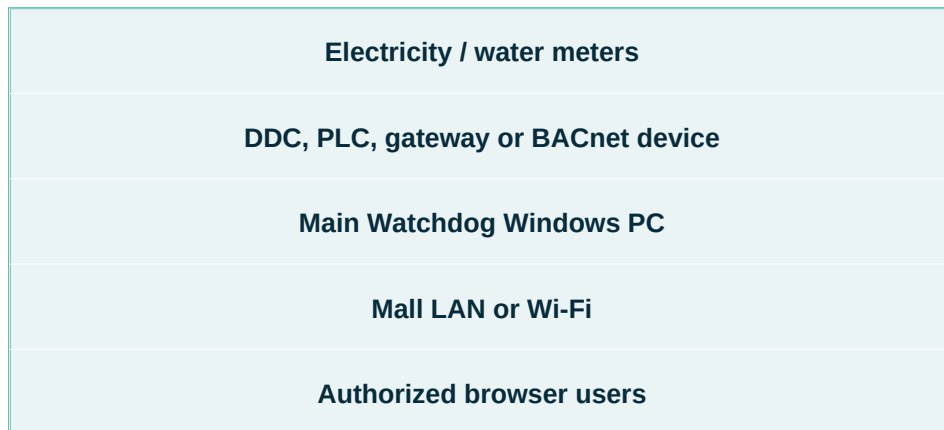
- 1 Open the Dashboard and review communication alarms.
- 1 Confirm the latest reading time for important meters.
- 1 Open Grid, Generator or Water and search for any meter requiring attention.
- 1 Review the Published invoices page after the scheduled publication day.
- 1 Report unexpected decreases, stale readings or failed communication before billing.

Keep the server running

The main Watchdog PC must remain powered on. Configure Windows not to sleep. Locking Windows with Win+L is acceptable when sleep is disabled; closing the browser does not stop the Windows service.

2. System architecture and access

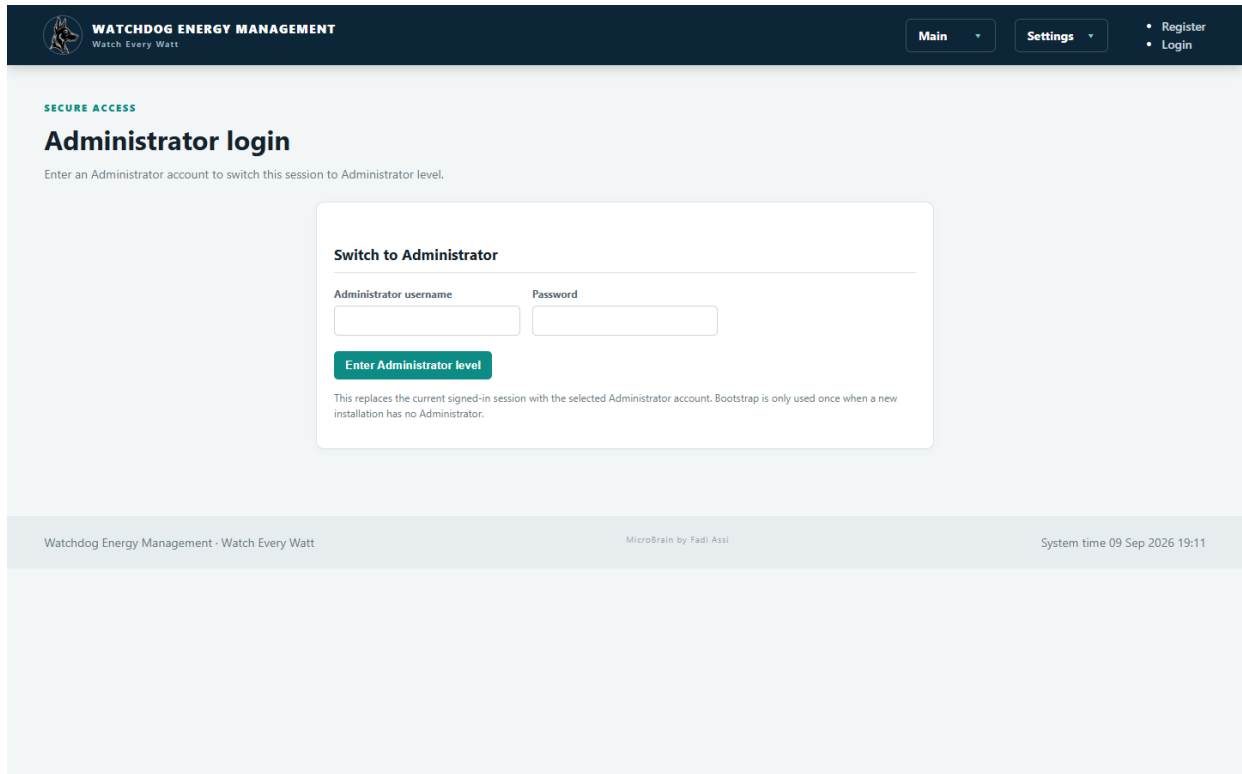
Meters communicate only with the main Watchdog PC through the configured controller or gateway. Other users connect to the Watchdog web application with a browser; they do not poll meters directly.



Local and network addresses

Field or status	Meaning
Main PC	http://localhost:5080
Another device	http://MAIN-PC-IP:5080, for example http://192.168.16.201:5080
Required firewall rule	Inbound TCP port 5080 on the main Watchdog PC, normally limited to the private/local network.
Database	One shared database on the main PC. Browser clients see the same meters, invoices and settings.

3. Signing in



The screenshot shows the Watchdog Energy Management web interface. The header is dark blue with the Watchdog logo and tagline 'Watch Every Watt' on the left. On the right, there are links for 'Main', 'Settings', 'Register', and 'Login'. The main content area is light gray and features a 'SECURE ACCESS' section titled 'Administrator login'. Below the title, it says 'Enter an Administrator account to switch this session to Administrator level.' A white box contains the 'Switch to Administrator' form with fields for 'Administrator username' and 'Password', and a green 'Enter Administrator level' button. A note below the button states: 'This replaces the current signed-in session with the selected Administrator account. Bootstrap is only used once when a new installation has no Administrator.' The footer is light gray and contains the text 'Watchdog Energy Management · Watch Every Watt', 'MicroBrain by Padi Assi', and 'System time 09 Sep 2026 19:11'.

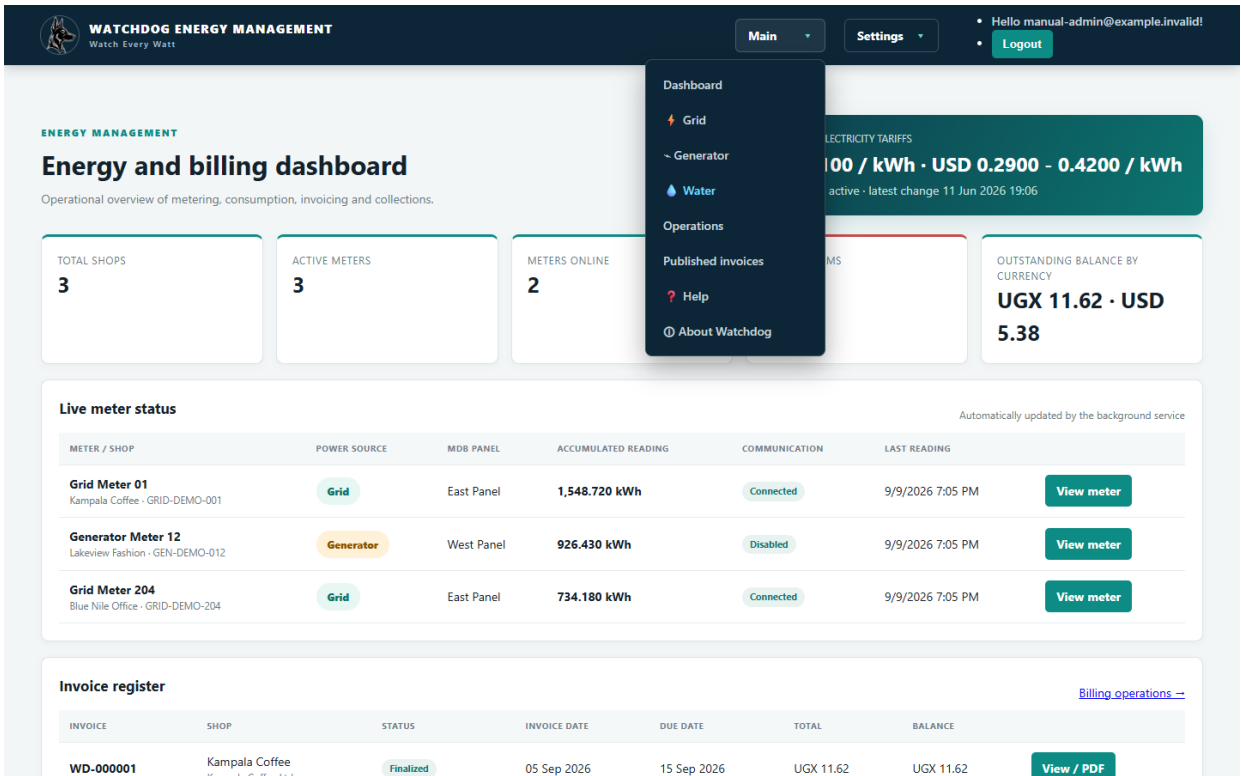
Administrator login screen. Standard users normally use the Sign in option in the header.

- 1 Open Watchdog in a supported browser.
- 2 Select **Login** or **Sign in** in the upper-right area.
- 3 Enter the username created by the Administrator. The username is not necessarily the contact email.
- 4 Enter the password and submit the form.
- 5 If administrator configuration is required, open **Settings - Administrator login** and enter an Administrator account.

Security

Do not share accounts. Five failed attempts can temporarily lock an account. Ask an Administrator for assistance; passwords are not displayed by Watchdog.

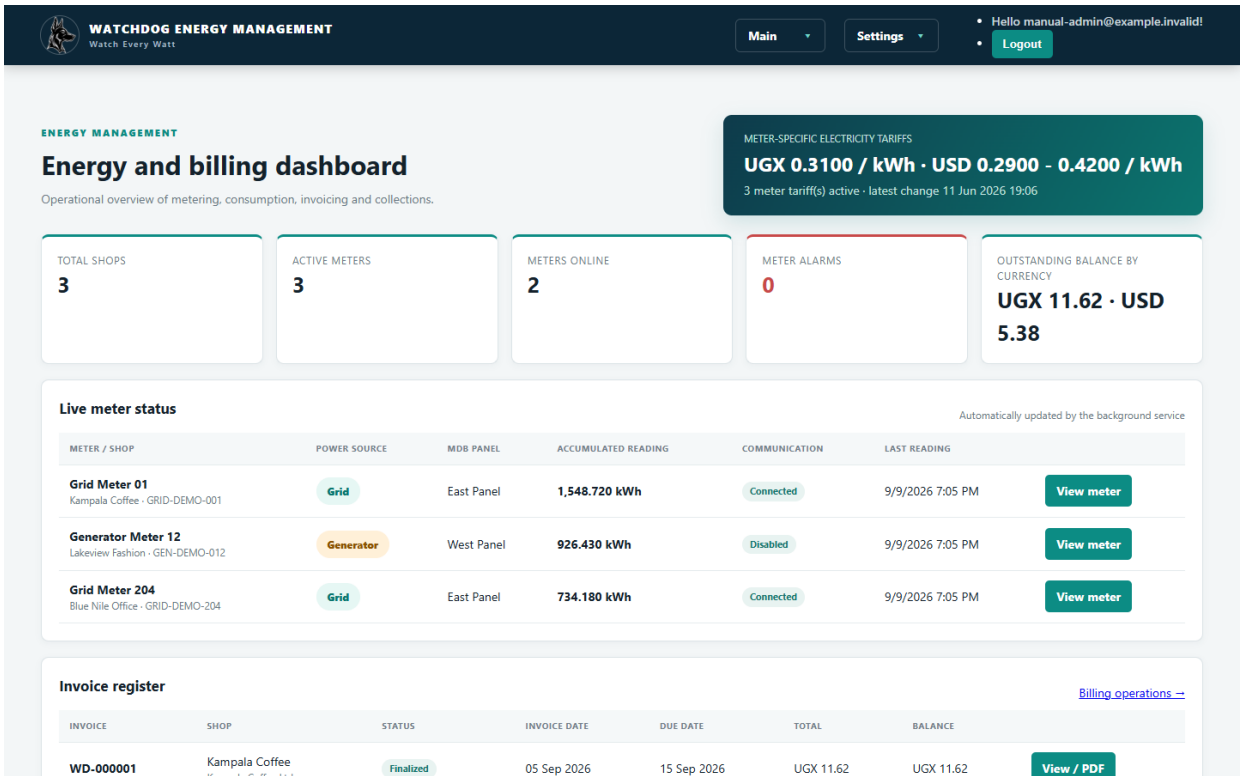
4. Navigation



The Main menu provides monitoring, Operations, invoices, Help and About. The Settings menu contains Administrator controls.

Field or status	Meaning
Dashboard	System overview, live electricity meters, current tariffs, invoice register and condition summary.
Grid	Electricity meters assigned to the Grid source.
Generator	Electricity meters assigned to the Generator source.
Water	Optional water meters. Visible only when Water metering and billing is enabled.
Operations	Tariff and publication schedule, controller/shop summaries, readings, CSV export and Administrator backup controls.
Published invoices	Searchable archive of finalized invoices and PDF access.
Help	Opens this bundled manual in a new browser tab. Use the browser PDF Download control to save a copy.
Settings	Administrator pages for shops, controllers, features, license, users, email, audit and backup.
About Watchdog	Displays software version and product information.

5. Dashboard



Dashboard with safe demonstration data.

Field or status	Meaning
Shops	Number of configured shop/customer records.
Active meters	Commissioned meters currently marked active.
Meters online	Meters with a healthy current communication status.
Meter alarms	Meters reporting a failed or unhealthy communication state.
Outstanding balance	Unpaid totals grouped by currency; currencies are never added together.
Tariff banner	Current meter-specific tariff range grouped by currency.
Accumulated reading	The meter's lifetime or continuously increasing register value, not the monthly consumption.
Last reading	Time of the most recent successful value stored by Watchdog.
View meter	Opens the selected meter's detailed consumption charts.

Reading versus consumption

The Dashboard shows accumulated meter readings. Monthly consumption and invoice cost are calculated from the difference between readings inside the billing period.

6. Grid and Generator monitoring

**WATCHDOG ENERGY MANAGEMENT**
Watch Every Watt

[Main](#) [Settings](#)

Hello manual-admin@example.invalid!

[Logout](#)

POWER SOURCE

Grid

Electricity meters supplied by the public utility grid.

2 meters

**Find a Grid meter**
Search by meter number, ID, name, customer, tenant, shop, room, or location.

[Search](#)

Grid meter register

METER	CUSTOMER / TENANT	SHOP / LOCATION	CONTROLLER	READING	COMMUNICATION	
Grid Meter 01 ID 1 - GRID-DEMO-001	Kampala Coffee Ltd Grace N.	SHOP-01 - Kampala Coffee Ground - East Wing	East DDC East Panel	1,548.720 kWh 9/9/2026 7:05 PM	Connected	View meter
Grid Meter 204 ID 4 - GRID-DEMO-204	Blue Nile Services Amina S.	ROOM-204 - Blue Nile Office Second - Office Block	East DDC East Panel	734.180 kWh 9/9/2026 7:05 PM	Connected	View meter

Watchdog Energy Management - Watch Every Watt

MicroBrain by Fadi Assi

System time 09 Sep 2026 19:11

Grid page: only electricity meters assigned to Grid are displayed.

Search for a meter

- 1 Open **Main - Grid** or **Main - Generator**.
- 2 Enter a meter ID, serial number, meter name, customer, tenant, shop number, floor, zone, address, controller or panel.
- 3 Select **Search**.
- 4 Select **Clear** to restore the full list.
- 5 Select **View meter** to open consumption charts.

Both pages refresh automatically. If no item matches, Watchdog displays 'No matching meter was found.'

**WATCHDOG ENERGY MANAGEMENT**
Watch Every Watt

[Main](#) [Settings](#)

Hello manual-admin@example.invalid!

[Logout](#)

POWER SOURCE

Generator

Electricity meters supplied by on-site generators.

1 meter

**Find a Generator meter**
Search by meter number, ID, name, customer, tenant, shop, room, or location.

[Search](#)

Generator meter register

METER	CUSTOMER / TENANT	SHOP / LOCATION	CONTROLLER	READING	COMMUNICATION	
Generator Meter 12 ID 2 - GEN-DEMO-012	Lakeview Retail Ltd Daniel K.	SHOP-12 - Lakeview Fashion First - Central Atrium	BACnet Energy Server West Panel	926.430 kWh 9/9/2026 7:05 PM	Disabled	View meter

Watchdog Energy Management - Watch Every Watt

MicroBrain by Fadi Assi

System time 09 Sep 2026 19:11

Generator page: only electricity meters assigned to Generator are displayed.

7. Water monitoring

The screenshot shows the 'Water' section of the Watchdog Energy Management interface. The header includes the company logo, name, and navigation links for 'Main' and 'Settings'. A user status bar shows 'Hello manual-admin@example.invalid!' and a 'Logout' button. The main content area is titled 'Water' with a subtitle 'Live accumulated-volume readings from M-Bus meters through Modbus gateways.' Below this is a search bar labeled 'Find a water meter' with a placeholder 'Search water meters...' and a 'Search' button. A table titled 'Water meters' displays one result: 'Water Meter 204' (Blue Nile Office - Blue Nile Services - WATER-DEMO-204). The table columns are 'METER / SHOP', 'GATEWAY', 'REGISTER', 'ACCUMULATED VOLUME', 'COMMUNICATION', and 'LAST READING'. The accumulated volume is '438.265 m³'. A 'View meter' button is next to the last reading '9/9/2026 7:04 PM'. The footer shows 'Watchdog Energy Management - Watch Every Watt', 'MicroBrain by Fadi Assi', and 'System time 09 Sep 2026 19:11'.

METER / SHOP	GATEWAY	REGISTER	ACCUMULATED VOLUME	COMMUNICATION	LAST READING
Water Meter 204 Blue Nile Office - Blue Nile Services - WATER-DEMO-204	Water Gateway 192.168.20.40:502	4100 UInt32 - LowHigh - x 0.001	438.265 m³	Connected	9/9/2026 7:04 PM

Water uses blue styling and shows accumulated volume in cubic metres (m3).

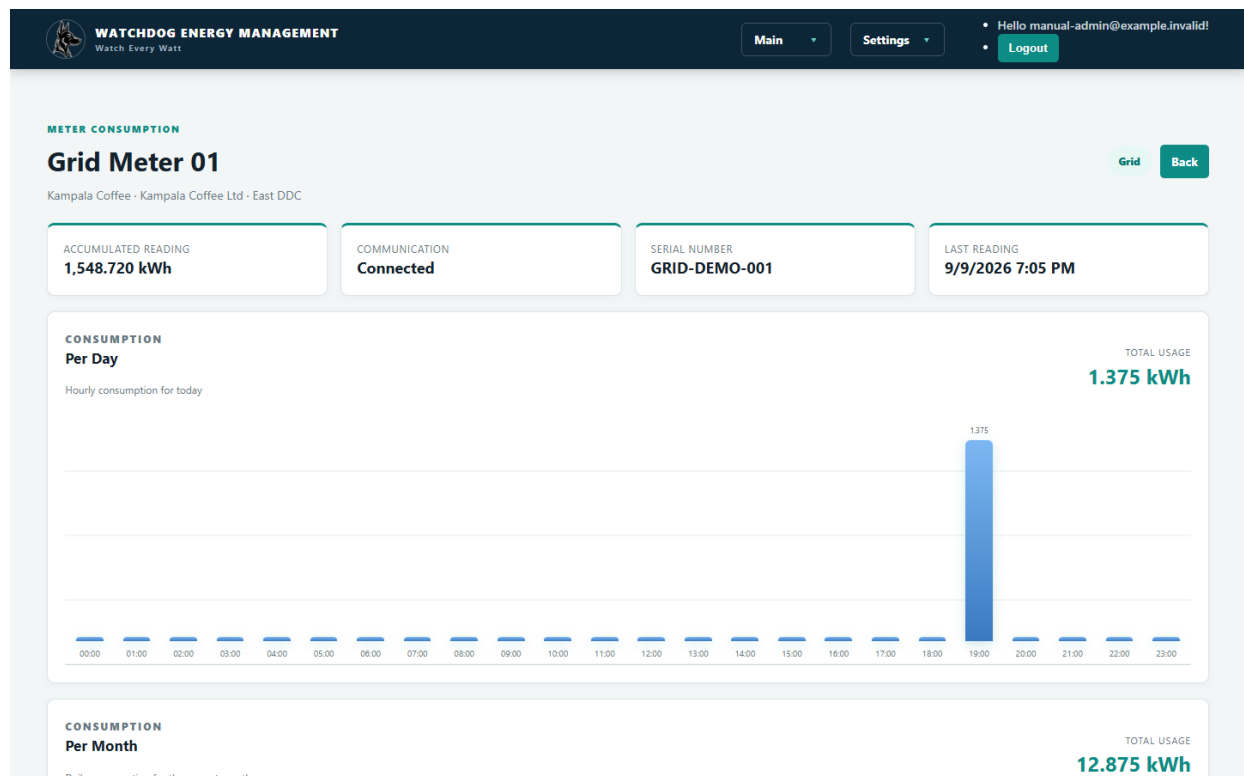
Water is optional. An Administrator enables it under **Settings - System features**. Watchdog reads the accumulated volume exposed by an M-Bus-to-Modbus gateway. The gateway may use Modbus TCP/IP or another supported Modbus path configured as a controller.

- 1 Open **Main - Water**.
- 2 Search by meter, customer, tenant, shop, room, location, controller or gateway.
- 3 Check accumulated volume, communication state and latest reading time.
- 4 Select **View meter** for daily, monthly and yearly volume charts.

Units

Electricity uses kWh. Water uses m3. Tariffs and invoice descriptions follow the meter utility type.

8. Meter detail and consumption charts



Meter detail page with summary cards and Per Day consumption chart.

Field or status	Meaning
Per Day	Hourly consumption for the current day.
Per Month	Daily consumption for the current month.
Per Year	Monthly consumption for the current year.
Bar tooltip	Hover over or focus a bar to see its period, consumption, and cost when a tariff is available.
Total usage	Sum of valid consumption differences in the displayed period.
No readings message	The chart will populate after enough valid cumulative readings are collected.
Lower reading	A value lower than the previous cumulative reading is marked for review and excluded from normal graph calculation.

Chart cost is based on the tariff effective at the time of each reading interval. The accumulated lifetime reading itself is not multiplied by the tariff.

9. Communication statuses and alarms

Field or status	Meaning
Connected / Online	A recent valid response was received.
Partial Data	BACnet communication is available but one or more optional points may be missing or unhealthy.
Awaiting poll	The meter is active and waiting for the first scheduled poll.
Not commissioned	The channel exists but is not activated for live polling.
Disabled	The meter or controller is disabled.
Stale	The latest BACnet success is older than the allowed polling window.
Failed / Configuration Error	Communication or configuration failed. Open controller/meter settings and review the last error.
Archived	The meter is hidden from live monitoring but its historical records are preserved.
Requires review	The reading is not safe for automatic billing until investigated.

First response to an alarm

- 1 Check that the physical meter, DDC, PLC or gateway has power.
- 2 Check Ethernet, Wi-Fi, RS-485 and USB converter connections.
- 3 Confirm the controller is enabled and the meter is commissioned.
- 4 Review COM port, IP address, ports, slave address, register or BACnet point mapping.
- 5 Use the controller Test function. Do not finalize billing from uncertain readings.

10. Shops, customers and tenants

The screenshot shows the 'Shop setup' page in the Watchdog Energy Management system. The header includes the logo, 'WATCHDOG ENERGY MANAGEMENT', and navigation links for 'Main' and 'Settings'. A user status bar shows 'Hello manual-admin@example.invalid!' and a 'Logout' button. The page is titled 'ADMINISTRATOR ONLY' and 'Shop setup'. A subtitle reads 'Create a shop before assigning meter channels.' A 'Controller setup' button is in the top right. The main form is titled 'Add shop' and contains several input fields: 'Shop number' (with 'SHOP-01' entered), 'Shop name', 'Tenant name', 'Floor', 'Zone', 'MDB panel', 'Contact person', 'Telephone', 'Email', 'Status' (a dropdown menu with 'Active' selected), 'Billing address', and 'Tax number'. There are also optional text areas for 'Notes' and 'Reason'. An 'Add shop' button is at the bottom of the form. Below the form is a section titled 'Configured shops'.

Administrator Shop setup page.

A shop is the billing customer/location record assigned to one or more meters. Create the shop before assigning meter channels.

- 1 Open **Settings - Shops**.
- 2 Enter a unique shop number and a clear shop name.
- 3 Enter tenant, floor, zone and panel information.
- 4 Enter contact person, telephone and customer email if available.
- 5 Choose Active, Vacant or Inactive.
- 6 Enter billing address and tax number when required.
- 7 Select **Add shop**. Use **Edit shop** later to update contact or billing information.

Email address

The customer email is used as the suggested invoice recipient. Confirm it before enabling automatic invoice email.

11. Controller setup overview



WATCHDOG ENERGY MANAGEMENT
Watch Every Watt

Main

Settings

Hello manual-admin@example.invalid!

Logout

ADMINISTRATOR ONLY

Controller communication

+ Add controller

Add controllers and select Modbus RTU (RS-485), Modbus TCP/IP (Ethernet), or BACnet/IP (UDP). These controls are hidden from all other roles.

CONTROLLER	COMMUNICATION	CONNECTION SETTINGS	DEVICE ADDRESS	CONDITION			
BACnet Energy Server West Panel	BACnet/IP	192.168.20.60:47808 - local 192.168.20.10	Instance 1201	Disabled	Edit / test	45 channels	Delete
East DDC East Panel	Modbus RTU	COM2 - 9600 - Even - 8/1	Slave / unit 1	Disabled	Edit / test	45 channels	Delete
Water Gateway Water Services	Modbus TCP/IP	192.168.20.40:502	Slave / unit 1	Disabled	Edit / test	45 channels	Delete

Watchdog Energy Management - Watch Every Watt

MicroBrain by Fadi Asri

System time 09 Sep 2026 19:11

Administrator controller list showing Modbus RTU, Modbus TCP/IP and BACnet/IP.

- 1 Open **Settings - Controller setup**.
- 2 Select **Add controller**.
- 3 Enter a name and panel/location.
- 4 Choose Modbus RTU, Modbus TCP/IP or BACnet/IP.
- 5 Complete the protocol-specific fields.
- 6 Save the controller, create meter channels, edit each meter, commission it and test communication.

A controller represents the communication endpoint used for one or more meters. Each controller supports up to 45 configured meter channels in the user interface.

12. Modbus RTU configuration

Field or status	Meaning
Windows port	COM number of the USB/RS-485 adapter, such as COM2.
Baud rate	Must match the DDC, PLC or meter gateway.
Parity / data bits / stop bits	Serial framing parameters. All devices on the bus must match.
Slave address	Modbus address of the target device, usually 1 to 247.
Polling interval	Seconds between reading cycles.
Timeout / retries	How long Watchdog waits and how many additional attempts it makes.
Starting register	First holding-register address for the meter value.
Data type	UInt16, Int16, UInt32, Int32, Float32 or UInt64, according to the register map.
Word order	HighLow or LowHigh for multi-register values.
Scaling factor	Multiplier that converts the raw register value into kWh or m3.

Address notation

Some manuals use 40001-style references while software libraries use zero-based addresses. Confirm the exact value with a live test; do not guess.

13. Modbus TCP/IP configuration

Field or status	Meaning
Device IP or host	Network address of the DDC, PLC or gateway.
Modbus TCP port	Normally 502 unless the gateway uses another port.
Unit address	Unit/slave identifier forwarded by the gateway.
Register mapping	Same data type, word order and scaling rules as Modbus RTU.
Water gateway	An M-Bus gateway can expose a water meter's accumulated volume through Modbus TCP/IP.

- 1 Confirm that the Watchdog PC and gateway are reachable on the same routed network.
- 2 Use a fixed IP address or stable DHCP reservation for the gateway.
- 3 Allow TCP port 502, or the configured custom port, through applicable network firewalls.
- 4 Test the first configured meter before commissioning all channels.

14. BACnet/IP configuration



WATCHDOG ENERGY MANAGEMENT
Watch Every Watt

Main

Settings

Hello manual-admin@example.invalid!

Logout

ADMINISTRATOR ONLY

Edit controller

Back to controllers

Configure, confirm and test communication before enabling collection.

1. Controller identity

Name

BACnet Energy Server

MDB panel

West Panel

Communication type

BACnet/IP (UDP)

☐ Enabled

2-3. Communication address

Device IP address or host name

192.168.20.60

BACnet UDP port

47808

Local BACnet network interface

Select the network connected to your meter

BACnet device instance

1201

Default BACnet/IP port is 47808.

Interfaces on the Watchdog server. No adapter is selected automatically.

4. Polling resilience

Polling interval (seconds)

15

Response timeout (ms)

2000

Retry count

2

5. BACnet connection and discovery

Discover BACnet Devices

Test BACnet Connection

BACnet/IP controller configuration and discovery controls.

Field or status	Meaning
Device IP or host	Target BACnet device address.
BACnet UDP port	Default 47808 (hex BAC0), unless the device uses another BACnet port.
Local BACnet interface	The specific network adapter/IP on the Watchdog PC connected to the BACnet network.
Device instance	BACnet Device object instance discovered or entered for the target.
Discover devices	Searches for reachable BACnet devices on the selected interface.
Test connection	Reads the device identity/communication path without changing billing data.
Browse objects	Lists device objects when the device permits Object_List browsing.

Network boundary

BACnet broadcast discovery may not cross routers or VLANs without a BACnet Broadcast Management Device (BBMD) or network configuration by the site integrator. Direct reads can still work when routing permits.

15. Creating meter channels

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• [Logout](#)

ADMINISTRATOR ONLY

East DDC — meter channels

Controller communication

2 of 45 configured. Create meter entries, then commission each meter.

Bulk-create meter channels

Configure sequential Modbus registers or BACnet objects according to the selected controller communication type.

Meter name prefix Meter	First meter serial number Enter the physical serial when available	Controller / gateway East DDC
Utility type Electricity	Electricity source Grid	Number of channels 43
Water readings are stored and billed in m³. Used only by electricity meters.		
First register address 4196	Register stride 2	Data type UInt32
Word order LowHigh	Scaling factor 0.01	Initial shop assignment ROOM-204 — Blue Nile Office
Reason (optional) 		
Generate channels		

Channel matrix

Administrator controls: Archive preserves history and hides the meter from live queries. Permanently delete removes the meter and all related readings, tariffs, invoices, and documents.

Bulk channel creation. The fields change according to the controller protocol.

- 1 Open the controller and select **45 channels**.
- 2 Choose the number of channels to add, up to the controller limit.
- 3 Enter a meter name prefix and, when available, the first physical serial number.
- 4 Select Electricity or Water. For electricity, select Grid or Generator.
- 5 For Modbus, enter first register, stride, data type, word order and scale. For BACnet, enter object type, first object instance and instance stride.
- 6 Choose an initial shop assignment and generate channels.
- 7 Open each generated meter, replace pending serial data, verify the mapping, select **Commissioned and active**, and save.

License capacity

Free mode allows up to five active meters. A signed commercial license raises the active-meter allowance. Existing data remains preserved if a license becomes invalid or over capacity.

16. Editing and commissioning a meter

Meter name Grid Meter 01	Physical meter serial number GRID-DEMO-001	Controller / gateway East DDC
Utility type Electricity	Electricity source Grid	First register address 4096
Select Water for an M-Bus meter exposed by a Modbus gateway.		
Used only when Utility type is Electricity.		
Data type UInt32	Word order LowHigh	Shop assignment SHOP-01 — Kampala Coffee
☒ Commissioned and active	Scaling factor 0.01	Pulse constant (electricity only) 1600
Use only the conversion specified by the meter/gateway object/register map.		
CT primary (electricity only) 1.00	CT secondary (electricity only) 1.00	☒ CT ratio already applied (electricity only)
Notes (optional) Safe demonstration data		
Reason (optional)		
<button>Save meter channel</button>		

Meter channel configuration for a Modbus electricity meter.

Field or status	Meaning
Meter name	Clear customer-facing name used on monitoring and invoices.
Serial number	Unique physical meter serial number.
Utility type	Electricity or Water.
Electricity source	Grid or Generator; used only for electricity.
Shop assignment	Customer/location receiving the invoice.
Commissioned and active	Allows live polling and counts toward license capacity.
Scale	Raw-value conversion from the manufacturer register map.
CT fields	Electricity transformer details. Mark CT already applied only when the PLC/meter output already includes it.
Notes / reason	Optional operational context; configuration changes are recorded in the audit trail.

17. BACnet meter points

**WATCHDOG ENERGY MANAGEMENT**
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• Hello manual-admin@example.invalid!
• [Logout](#)

BACNET COMMISSIONING

Generator Meter 12 — BACnet points

Meter settings

Map numerical objects to meter measurements. Billing uses only the explicitly confirmed cumulative import-energy point.

Device diagnostics

Local interface **192.168.20.10**

Target **192.168.20.60:47808**

Device Instance **1201**

Status **Disabled**

Last polling attempt 9/9/2026 7:05:56 PM +03:00

Last response 9/9/2026 7:05:56 PM +03:00

Consecutive failures 0

Last error

[Edit network](#) / [Test BACnet Connection](#) / [Discover BACnet Devices](#)

Configured points

Add Point

POINT / MEASUREMENT	OBJECT / PROPERTY	LAST VALUE	QUALITY / LAST SUCCESS	
Total Import Energy Unmapped — billing confirmed	AnalogValue41 / 85	926.43 kWh	Stale 9/9/2026 7:05:56 PM +03:00	Edit / Test Read Delete

Add Point

Point Name	Watchdog Measurement	BACnet Object Type
<i>BACnet point diagnostics, configured points and mapping controls.</i>		

BACnet meters use explicit point mappings. Billing is allowed only from a point deliberately confirmed as cumulative Total Import Energy in kWh.

- 1 Open the BACnet meter and select **BACnet Points**.
- 2 Use **Browse BACnet Objects**, or enter the object manually if browsing is restricted.
- 3 Enter point name, Watchdog measurement, object type, object instance and property identifier. Present_Value is property 85.
- 4 Enter the final engineering unit, scale and offset.
- 5 Select **Test Read** and compare the result with the physical meter.
- 6 For billing, confirm only the cumulative Total Import Energy kWh point.
- 7 Save the point and verify Good quality and a recent timestamp.

Do not confirm demand or power

kW, demand, current, voltage and interval energy are not cumulative billing counters. Confirm only the physical meter's cumulative import-energy value for electricity billing.

18. Tariffs and monthly invoice publication

Tariff and monthly invoice publication

Meter

Select meter

Tariff price

Currency

UGX

Invoice publication day

Select day 1-7

Payment due after (days)

10

Reason (optional)

The tariff starts immediately for the selected meter. Every month, invoices for the previous completed month are published on the selected day.

Save all settings

Current schedule: day 5 of every month · next publication 14 Sep 2026 · payment due after 10 days

Previous tariff and schedule changes are available only in [Audit trail](#)

Tariff and monthly invoice publication controls on Operations.

- 1 Open **Main - Operations** and go to **Tariffs & invoice day**.
- 2 Select one meter.
- 3 Enter its tariff price and ISO currency code.
- 4 Select publication day 1 through 7.
- 5 Enter how many days after publication payment becomes due.
- 6 Optionally enter a reason, then select **Save all settings**.
- 7 Repeat for every meter requiring a separate tariff.

The tariff begins immediately for the selected meter. The global monthly schedule publishes invoices for the previous completed calendar month shortly after midnight on the selected day. Watchdog must be running, valid readings must exist for the period, and the meter must have a usable tariff.

One schedule, individual tariffs

The publication day and due-day setting are system-wide. Tariff price and currency are stored per meter.

19. How invoice calculation works

Field or status	Meaning
Opening reading	First accepted cumulative reading used for the billing period.
Closing reading	Last accepted cumulative reading used for the billing period.
Consumption	Closing reading minus opening reading.
Tariff	Price per kWh or per m3 effective for that meter.
Utility charge	Consumption multiplied by tariff.
Previous balance	Outstanding amount carried according to the account-balance rules.
Total and balance	Invoice charge and amount still unpaid, shown in the invoice currency.
Finalized / published	Locked customer invoice available in Published invoices and PDF form.
Partially paid / Paid / Overdue	Status derived from payment and due-date information already recorded by the system.

No invoice created

Common causes are no valid readings in the completed month, no tariff for the meter, an unconfirmed BACnet billing point, an inactive meter, or Watchdog not running at the scheduled time.

20. Published invoices

**WATCHDOG ENERGY MANAGEMENT**
Watch Every Watt

[Main](#) [Settings](#)

• Hello manual-admin@example.invalid!
• [Logout](#)

BILLING ARCHIVE

Published invoices

Every finalized monthly invoice is saved here for viewing and PDF download.

 **Find an invoice**
Search by day, month, year, invoice number, shop, tenant or meter.

Example: 01, September 2026, SHOP-01 or tenant name [Search](#)

Invoice archive 2 result(s)

INVOICE	SHOP / METER	PUBLISHED	DUE DATE	STATUS	TOTAL	BALANCE	
WD-000002	Lakeview Fashion Lakeview Retail Ltd - Generator Meter 12 - Generator	05 Sep 2026	15 Sep 2026	PartiallyPaid	USD 9.38	USD 5.38	View / PDF
WD-000001	Kampala Coffee Kampala Coffee Ltd - Grid Meter 01 - Grid	05 Sep 2026	15 Sep 2026	Finalized	UGX 11.62	UGX 11.62	View / PDF

Watchdog Energy Management · Watch Every Watt MicroBrain by Fadi Asli System time 09 Sep 2026 19:11

Published invoices archive and search banner.

- 1 Open **Main - Published invoices**.
- 2 Search by day, month, year, shop name, tenant, meter or invoice number.
- 3 Select **Clear** to remove the search.
- 4 Select **View / PDF** to review an invoice.
- 5 Confirm the customer, meter, dates, readings, tariff, charges and balance before distribution.

Draft or cancelled invoices are not customer-ready. Finalized, partially paid, paid and overdue statuses describe the invoice's current state.

21. Invoice details, PDF and email

**WATCHDOG ENERGY MANAGEMENT**
Watch Every Watt

[Main](#) [Settings](#)

Hello manual-admin@example.invalid!

[Logout](#)

ELECTRICITY INVOICE

WD-000002

Download PDF Print preview

Lakeview Fashion - PartiallyPaid

Email PDF invoice

SMTP invoice email is disabled. An Administrator can configure and activate it in [Settings -> Invoice email](#).

Mall Utility Metering & Billing

Billed to

Lakeview Fashion
Shop SHOP-12
Tenant: Lakeview Retail Ltd

Meter

Generator Meter 12
GEN-DEMO-012
Utility: Electricity
Power source: Generator
West Panel

Due 15 Sep 2026

PREVIOUS READING	CURRENT READING	CONSUMPTION	TARIFF
904.10 kWh	926.43 kWh	22.33 kWh	USD 0.4200 / kWh
Electricity charge			USD 9.38
Fixed and additional charges			USD 0.00
Tax			USD 0.00
Discount			- USD 0.00
Previous balance			USD 0.00

Invoice detail page with Download PDF, Print preview and optional email controls.

- 1 Select **Download PDF** to create the official invoice file.
- 2 Select **Print preview** to use the browser print workflow.
- 3 When SMTP is enabled and your role permits it, confirm the recipient and select **Send PDF by email**.
- 4 Verify the success message. If sending fails, the invoice remains stored and can be retried.

Customer address

Automatic delivery uses the configured customer/shop email. Keep addresses current and test SMTP before enabling global automatic delivery.

22. Invoice email settings



WATCHDOG ENERGY MANAGEMENT
Watch Every Watt

Main

Settings

Hello manual-admin@example.invalid!

Logout

ADMINISTRATOR ONLY

Invoice email

Optionally send published PDF invoices through the customer's SMTP provider.

SMTP configuration

Disabled by default. The password is encrypted for this Windows installation and is never displayed again. After a successful test, the settings and password are saved automatically.

☐ Enable PDF invoice email

☐ Automatically email every newly published invoice

Port

587

SMTP password or app password

Enter SMTP password

Test recipient

billing@example.com

Standard email message

Please find your utility invoice attached. Thank you for your business.

SMTP server

smtp.example.com

SMTP username

Sender email

billing@example.com

Sender name

Watchdog Energy Management

Use SSL/TLS (STARTTLS)

☒

Save SMTP settings

Save and send test email

Administrator SMTP configuration. Demonstration fields contain no credentials.

Field or status	Meaning
Enable PDF invoice email	Master switch for all invoice email sending.
Automatically email	Queues every newly published invoice for customer delivery.
SMTP server / port	Mail provider host and connection port.
SSL/TLS	Enables STARTTLS according to the provider configuration.
Username / password	SMTP credentials or provider app password. The password is encrypted for this Windows installation and is not displayed again.
Sender email / name	Identity displayed to customers.
Test recipient	Address used only for the test message.
Standard email message	Reusable text inserted after 'Dear customer' and before invoice details.
Save and send test	Validates and saves the current configuration, then sends a test message.

Global enable/disable

Turn on both switches for automatic delivery. Turn off automatic delivery to keep manual sending only. Turn off PDF invoice email to disable all sending.

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23. Reports and reading history

7/29/2026 5:06 PM	Grid Meter 204 Electricity	72225	722.254 kWh	Automatic	Good
7/28/2026 4:06 PM	Grid Meter 204 Electricity	72166	721.666 kWh	Automatic	Good
7/27/2026 3:06 PM	Grid Meter 204 Electricity	72102	721.029 kWh	Automatic	Good
7/26/2026 7:06 PM	Grid Meter 204 Electricity	72053	720.539 kWh	Automatic	Good
9/9/2026 7:06 PM	Water Meter 204 Water	44539	445.399 m ³	Automatic	Good
9/8/2026 6:06 PM	Water Meter 204 Water	44499	444.996 m ³	Automatic	Good
9/7/2026 5:06 PM	Water Meter 204 Water	44468	444.686 m ³	Automatic	Good
9/6/2026 4:06 PM	Water Meter 204 Water	44434	444.345 m ³	Automatic	Good
9/5/2026 3:06 PM	Water Meter 204 Water	44397	443.973 m ³	Automatic	Good
9/4/2026 7:06 PM	Water Meter 204 Water	44357	443.570 m ³	Automatic	Good
9/3/2026 6:06 PM	Water Meter 204 Water	44326	443.260 m ³	Automatic	Good
9/2/2026 5:06 PM	Water Meter 204 Water	44291	442.919 m ³	Automatic	Good
9/1/2026 4:06 PM	Water Meter 204 Water	44254	442.547 m ³	Automatic	Good
8/31/2026 3:06 PM	Water Meter 204 Water	44214	442.144 m ³	Automatic	Good
8/30/2026 7:06 PM	Water Meter 204 Water	44183	441.834 m ³	Automatic	Good

Operations reading history showing stored meter values, source and quality.

Watchdog's current reporting tools are the meter consumption charts, invoice PDFs, invoice archive search and the **Export readings CSV** command. There is no separate report designer in v4.2.0.

- 1 Open **Main - Operations**.
- 2 Select **Export readings CSV** to download stored reading data.
- 3 Open a meter to review hourly, daily and monthly consumption charts.
- 4 Use Published invoices for financial document history.
- 5 Use the audit trail for configuration and billing actions.

Field or status	Meaning
Raw	Raw stored value when available.
Accumulated	Converted cumulative kWh or m3 value.
Source	Automatic, Manual, Simulated or Imported.
Quality	Good or a warning/review state.
Used for billing	Internal indication that a reading participated in invoice calculation.

24. Backup and restore

Backup and restore

Backup folder: C:\Users\Fady\Documents\Codex\2026-08-25\files-pasted-by-the-user-build\manual-demo-data\Backups

Create new backup

BACKUP FILE	CREATED	SIZE	
watch-dog-em-auto-20260909.db	09 Sep 2026 19:01:38	0.21 MB	Restore

Administrator Backup and restore section.

- 1 Open **Settings - Backup & restore**, or Operations and scroll to Backup and restore.
- 2 Select **Create new backup**.
- 3 Confirm that a dated backup file appears with a reasonable size.
- 4 Copy important backups to secure storage according to the customer's policy.
- 5 To restore, choose the correct backup and select **Restore**.
- 6 Read the warning carefully. Restore replaces current database content; verify the result after Watchdog restarts.

Recovery test

A backup is not proven until it has been restored successfully in a controlled test. Never test restoration on the only live production database.

25. System features

The screenshot shows the 'System features' page in the Watchdog Energy Management interface. The page is titled 'ADMINISTRATOR ONLY' and 'System features'. Below the title, it says 'Enable only the utility services installed at this property.' The main content area is titled 'Water metering and billing' and contains a checkbox labeled 'Enable Water pages, water-meter configuration and water billing'. Below the checkbox, there is a note: 'Water meters are read from the accumulated-volume registers exposed by an M-Bus-to-Modbus TCP gateway. Watchdog Energy Management records and bills the gateway value; it does not calculate water volume.' At the bottom of the content area is a 'Save feature settings' button. The footer of the page shows 'Watchdog Energy Management - Watch Every Watt', 'MicroBrain by Fadi Assi', and 'System time 09 Sep 2026 19:11'.

WATCHDOG ENERGY MANAGEMENT
Watch Every Watt

Main Settings

• Hello manual-admin@example.invalid!
• Logout

ADMINISTRATOR ONLY

System features

Enable only the utility services installed at this property.

Water metering and billing

☒ Enable Water pages, water-meter configuration and water billing

Water meters are read from the accumulated-volume registers exposed by an M-Bus-to-Modbus TCP gateway. Watchdog Energy Management records and bills the gateway value; it does not calculate water volume.

Save feature settings

Watchdog Energy Management - Watch Every Watt

MicroBrain by Fadi Assi

System time 09 Sep 2026 19:11

Administrator Water feature switch.

Water pages and water-meter configuration are disabled by default. Enable the feature only when the property has water meters and a verified accumulated-volume communication path.

- 1 Open **Settings - System features**.
- 2 Select **Enable Water pages, water-meter configuration and water billing**.
- 3 Select **Save feature settings**.
- 4 Confirm that Water appears in the Main menu.
- 5 Configure a water shop/meter/gateway and verify m3 readings before assigning a tariff.

26. Users and permissions

WATCHDOG ENERGY MANAGEMENT
Watch Every Watt

Main Settings

Hello manual-admin@example.invalid!
Logout

ADMINISTRATOR ONLY

Users & Access

Create user accounts and assign one access level to each user.

Add new user

The username and password are used to sign in. Email is stored only as contact information.

Username:

Email: (Information only (optional))

Access level:

Temporary password:

Confirm password:

Create user

User access levels

USERNAME	EMAIL INFORMATION	CURRENT ACCESS	NEW ACCESS LEVEL
manual-admin@example.invalid Current account	manual-admin@example.invalid	Administrator	Administrator

Save access **Delete**

Access level permissions

ACCESS LEVEL	PERMISSION
--------------	------------

Administrator Users & Access page.

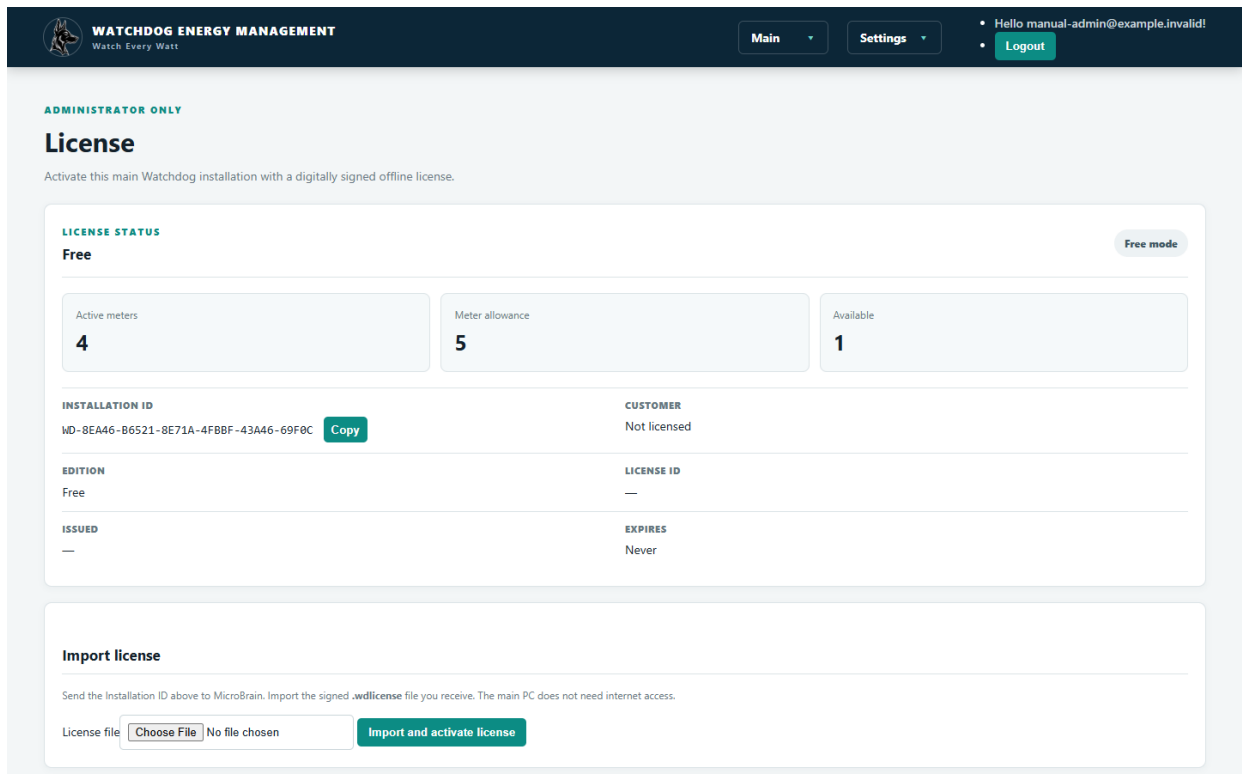
Field or status	Meaning
Administrator	Full configuration, controllers, meters, shops, billing, users, audit, licensing, email, backup and restore.
Billing Manager	Create, finalize and manage billing periods and invoices where those controls are available.
Operator	Operational pages and meter monitoring without Administrator configuration.
Viewer	Read-only access to permitted dashboards and operational information.

Create a user

- 1 Open **Settings - Users & Access**.
- 2 Enter a unique username. Allowed characters are letters, numbers, dots, underscores and hyphens.
- 3 Optionally enter an email for contact information.
- 4 Choose one access level.
- 5 Enter and confirm a temporary password of at least 10 characters.
- 6 Select **Create user** and give the credentials securely to the user.

Administrators can change access or delete another user. The current signed-in account cannot delete itself, and the last Administrator cannot be removed or demoted.

27. License activation



WATCHDOG ENERGY MANAGEMENT
Watch Every Watt

Main Settings

Hello manual-admin@example.invalid!
Logout

ADMINISTRATOR ONLY

License

Activate this main Watchdog installation with a digitally signed offline license.

LICENSE STATUS Free Free mode

Active meters 4	Meter allowance 5	Available 1
---------------------------	-----------------------------	-----------------------

INSTALLATION ID WD-8EA46-B6521-8E71A-4FBBF-43A46-69F0C Copy	CUSTOMER Not licensed
EDITION Free	LICENSE ID —
ISSUED —	EXPIRES Never

Import license

Send the Installation ID above to MicroBrain. Import the signed .wdlicense file you receive. The main PC does not need internet access.

License file Choose File No file chosen Import and activate license

Administrator License page. Installation IDs are unique to each main PC.

- 1 Open **Settings - License**.
- 2 Copy the Installation ID and send it to MicroBrain through the agreed secure business channel.
- 3 Receive the signed .wdlicense file.
- 4 Choose the file and select **Import and activate license**.
- 5 Confirm Commercial license, customer, edition and meter allowance.

Without a valid commercial license, free mode permits up to five active meters. A license is bound to the Installation ID and cannot simply be copied to another PC.

Private key

Customers never receive the MicroBrain private signing key. Only the signed .wdlicense file is imported into Watchdog.

28. Audit trail

The screenshot shows the 'Audit trail' page in the Watchdog Energy Management interface. The page is titled 'ADMINISTRATOR ONLY' and 'Audit trail'. Below the title, it states 'Protected history of configuration, billing, tariff, backup, and deletion activity.' A table displays the audit history with columns: TIMESTAMP, USER, ACTION, ENTITY, OLD VALUE, NEW VALUE, and REASON. The table contains one entry: '09 Sep 2026 19:06:56', 'Administrator', 'Demo configuration prepared', 'System 1', 'Safe demonstration data', and 'Customer manual screenshots'. A 'Back to operations' button is located in the top right corner. The footer of the page shows 'Watchdog Energy Management - Watch Every Watt', 'MicroBrain by Fadi Assi', and 'System time 09 Sep 2026 19:11'.

TIMESTAMP	USER	ACTION	ENTITY	OLD VALUE	NEW VALUE	REASON
09 Sep 2026 19:06:56	Administrator	Demo configuration prepared	System 1		Safe demonstration data	Customer manual screenshots

Administrator audit history.

The audit trail records important configuration, tariff, billing, backup, user and deletion activity with time, user, entity and reason/context. It is intended for accountability and troubleshooting.

- 1 Open **Settings - Audit trail**.
- 2 Locate the event by timestamp and action.
- 3 Review who performed it and which entity was affected.
- 4 Use audit evidence together with logs and backups when investigating an unexpected change.

29. Administrator maintenance checklist

Daily

- 1 Confirm Watchdog is running and Dashboard opens.
- 1 Review communication alarms and stale readings.
- 1 Confirm scheduled invoice/email errors are not present.

Monthly - before publication

- 1 Confirm each active meter has valid readings for the completed month.
- 1 Confirm meter-specific tariff and currency.
- 1 Check customer names, billing details and email addresses.
- 1 Confirm publication day and payment due days.
- 1 Create a fresh backup.

Monthly - after publication

- 1 Open Published invoices and verify the expected count.
- 1 Spot-check opening/closing readings, consumption and tariff.
- 1 Verify PDF output and automatic email results.
- 1 Investigate meters for which no invoice was created.

Quarterly

- 1 Test backup restoration on a non-production copy.
- 1 Review user accounts and access levels.
- 1 Review Windows sleep, service startup, disk space and network/firewall settings.
- 1 Compare selected Watchdog readings with physical meter displays.

30. Troubleshooting

Field or status	Meaning
Page cannot be reached on main PC	Confirm the Watchdog Windows service is running. Try http://localhost:5080 . Restart the service or PC if authorized.
Second PC cannot connect	Confirm the main PC IP, same LAN/Wi-Fi, private network profile and inbound TCP 5080 firewall rule. Test http://MAIN-PC-IP:5080 .
Not commissioned	Edit the meter, verify serial/mapping/shop, select Commissioned and active, save, then test the controller.
Modbus RTU failed	Check COM port, USB adapter, RS-485 polarity, baud/parity/data/stop bits, slave address and register map.
Modbus TCP failed	Check IP, port, unit address, gateway power, cable/VLAN and firewall/routing.
BACnet discovery empty	Select the correct local adapter. Confirm UDP port, subnet/VLAN, broadcast path and BBMD needs. Try direct device test.
BACnet value wrong	Verify object type/instance/property, unit, scale and offset against the physical meter.
Reading stale	Check poll interval, controller availability and last success. Do not use stale data as a fresh reading.
No invoice	Check active meter, valid period readings, tariff, schedule, billing-confirmed BACnet cumulative point and that Watchdog was running.
Email password required	Enter the SMTP/app password and use Save SMTP settings or Save and send test. Later saves can leave it blank to retain the encrypted saved value.
Email test fails	Confirm host, port, TLS, username, app password, sender permission, internet/DNS and provider security rules.
Download appears stuck	Wait for browser security scanning, use the ZIP release asset, verify free disk space and check the browser download panel without bypassing warnings.

31. Frequently asked questions

Does locking Windows stop Watchdog?

No. Win+L only locks the desktop. The Windows service should continue. Sleep, hibernation, power loss, forced updates or a stopped service can interrupt it.

Can several users view the system?

Yes. Authorized browser clients on the same LAN use the one main server and database. They do not create extra meter polling services.

Can each meter have a different tariff or currency?

Yes. Price and currency are meter-specific. Outstanding balances remain grouped by currency.

Does tariff date/time schedule the invoice?

No. In v4.2.0 the tariff becomes effective immediately when saved. Invoice publication uses the separate monthly day setting, from day 1 through day 7.

Can Water be hidden?

Yes. Disable it under System features when the property does not use water metering.

Can I delete a meter permanently?

Administrator controls support archive and permanent deletion. Archive preserves history. Permanent deletion removes dependent readings, tariffs, invoices and payments and requires explicit confirmation. Back up first.

Does Watchdog calculate electrical energy inside the PLC?

No. It reads the cumulative value supplied by the meter/controller and applies billing rules. The physical metering system remains responsible for measurement.

32. Glossary

Field or status	Meaning
Accumulated reading	Continuously increasing total from the meter, such as lifetime kWh or m3.
BACnet/IP	Building automation protocol transported over IP using UDP.
BACnet object	Addressed data object such as Analog Value with an instance number.
BBMD	BACnet Broadcast Management Device used to distribute BACnet broadcasts across IP subnets.
Billing period	Date range used to calculate consumption.
Commissioned	Verified and enabled for live operation.
Controller	DDC, PLC, gateway or BACnet device through which Watchdog obtains meter data.
CT	Current transformer used in electricity measurement.
CSV	Comma-separated file for spreadsheet/report analysis.
DDC	Direct Digital Controller.
Grid	Public utility electricity source.
Generator	On-site generated electricity source.
Invoice publication day	Configured day 1-7 when the previous completed month is invoiced.
M-Bus	Meter-Bus protocol commonly used by water and heat meters.
Modbus RTU	Serial Modbus protocol, commonly over RS-485.
Modbus TCP/IP	Modbus communication over Ethernet/IP.
Present_Value	BACnet property 85, commonly holding the current object value.
Scaling factor	Multiplier applied to a raw value to obtain the final engineering unit.
SMTP	Email delivery protocol used to send invoice PDFs.
Stale	Last successful value is older than the permitted polling window.
Tariff	Meter-specific price per kWh or m3.
Unit address	Modbus slave identifier, including gateway-forwarded devices.
Word order	Order of 16-bit words in multi-register Modbus values.

33. Support information

When requesting support, provide the software version shown in **About Watchdog**, the affected controller and meter names, the visible status/error, the approximate time, and whether other meters on the same controller are working.

Never send secrets

Do not send passwords, SMTP app passwords, private license signing keys or unrestricted customer databases by ordinary email. Use the approved secure support method.

Files and diagnostics

Field or status	Meaning
Database and configuration	Stored on the main Watchdog PC in the configured application data directory.
Backups	Displayed in Operations - Backup and restore.
Logs	Daily application logs in the Watchdog data Logs folder.
Version	Main - About Watchdog.
Manual	Main - Help. The PDF is bundled locally and can be downloaded from the browser PDF viewer.

End of Watchdog Energy Management Customer User Manual - Version 4.2.0